Mohammad Tahsin

Kingsville, Texas | +1 361 228 7268 | mohammad.tahsin@students.tamuk.edu
Website | Google Scholar | Linkedin

Research Profile

PhD student in Engineering with a Computer Science concentration at Texas A&M University-Kingsville and Graduate Research Assistant in the AI-Based Autonomous System Research Lab. Research spans machine learning for biomedical signals, machine learning applications, explainable AI, cybersecurity, etc.

Research Interests

Biomedical signal analysis and AI for healthcare • Explainable machine learning • Conformal Risk Prediction • Cybersecurity and anomaly detection • Deep Learning

Education

Texas A&M University-Kingsville, Kingsville, TX Aug. 2026–Present
PhD in Engineering, Computer Science concentration

Texas A&M University-Kingsville, Kingsville, TX Jan. 2025–Aug. 2026
MS in Computer Science

BRAC University, Dhaka, Bangladesh May 2017–Feb. 2022
BS in Computer Science

Research Experience

Graduate Research Assistant | AI-Based Autonomous System Research Lab May 2025–Present
Texas A&M University-Kingsville, Kingsville, TX

Current Project

- Zero-Day Intrusion Detection in Consumer Healthcare IoMT with Conformal Rejection (Under review in IEEE Transaction Series)
- AI-Driven Sensor Placement and Wireless Sensor Deployment: A PRISMA-Guided Systematic Review of Optimization Methods, Validation Maturity, and Benchmarking (Under review in MDPI journal)
- CGLGCTNet: A Lightweight Cross-Gated Local Global CNN Transformer Network for Plant Disease Classification (Under review in IEEE journal).
- FetalSANet: A Lightweight, Explainable Spatial-Attention MobileNetV2 Framework for Fetal Ultrasound Plane Classification (Under review in MDPI journal)
- Data-Local Soft-Voting Ensemble for Intrusion Detection in Consumer IoT Sensor-Gateway Networks (Under review in IEEE conference)
- Explainable Evasive PDF Malware Screening with TabPFN for Consumer Applications (Under review in IEEE conference)
- Systematic Evaluation and Explainability Analysis of Transfer Learning Models for Maize Leaf Disease Classification (Under review in IEEE conference)
- Interpretable Dual-Branch ConvNeXt–Transformer Fusion for Fine-Grained Acute Lymphoblastic Leukemia Classification (Under review in IEEE conference)
- PneuShield-FL: Differentially Private Federated Pneumonia Screening with Deep CXR Features and Explainable AI (Under review in IEEE conference)
- SafeSNM-QFed: A Leakage-Safe Stacked CNN–BFO Framework for Brain Tumor MRI Classification with Quantum and Federated Learning Extensions (Under preparation)
- Energy-Regularized Latent Synthetic Exposure for Open-Set Recognition of Clinical Tabular Data (Under review in IEEE journal)

Publications

2026

M. W. Rahman, H. Y. Adarbah, M. R. Khan, K. Roksana, M. S. Islam, **Tahsin, Mohammad Sadman**, and A. Noore, "FedQDFU: Non-IID Federated Quantum Learning for Diabetic Foot Ulcer Classification from Deep Image Features," *AI*, vol. 7, no. 10, 2026, doi: 10.3390/ai7100415

Mohammad Sadman Tahsin, H. Y. Adarbah, and A. Noore, "X-Beat: An Explainable Framework for ECG Image Classification," 2026, arXiv preprint, arXiv:2609.22631

S. Ahmed, N. Firoz, **Tahsin, Mohammad Sadman**, and D. Akpuluma, "Forecasting Carbon Footprint: Temporal Fusion Transformer Modeling for Predicting CO2 Emissions in Bangladesh," in *Innovations in Cybersecurity and Data Science*, S. M. Basha, H. Taherdoost, and N. Bačanin-Džakula, Eds. Cham: Springer Nature Switzerland, 2026, pp. 387–401, doi: 10.1007/978-3-032-12243-8_26

Tahsin, Mohammad Sadman, H. Y. Adarbah, and A. Noore, "A Lightweight Channel-Attention CNN for Robust Beat-Level Arrhythmia Detection," *IEEE Access*, vol. 14, pp. 62 803–62 817, 2026, doi: 10.1109/ACCESS.2026.3686810

Tahsin, Mohammad Sadman, H. Y. Adarbah, and A. Noore, "Smart-Home Sensor Placement Decision Support via Hybrid Model with GRU Energy Forecasting," in *2026 IEEE International Conference on Consumer Electronics (ICCE)*, 2026, pp. 1–6, doi: 10.1109/ICCE67443.2026.11449749

2024

N. Firoz, **Tahsin, Mohammad Sadman**, and M. Z. Hossain, "Autoencoder-Driven Anomaly Detection for Efficient Malware Identification," in *2024 Second International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI)*, 2024, pp. 1–8, doi: 10.1109/ICoICI62503.2024.10696300

Mohammad Sadman Tahsin, S. Abdullah, M. A. Karim, M. U. Ahmed, F. Tafannum, and M. Y. Ara, "A comparative study on data mining models for weather forecasting: A case study on Chittagong, Bangladesh," *Natural Hazards Research*, vol. 4, no. 2, pp. 295–303, 2024, doi: 10.1016/j.nhres.2023.12.014

2023

Tahsin, Mohammad Sadman, M. Al Karim, M. U. Ahmed, F. Tafannum, and N. Firoz, "An integrated approach for diabetes detection using fisher score feature selection and capsule network," *Journal of Computer Science and Engineering (JCSE)*, vol. 4, no. 2, pp. 61–77, 2023

G. Cano-Verdugo, J. Owoyemi, A. Khan, **Tahsin, Mohammad Sadman**, D. N. Shukurullaevna *et al.*, "Leveraging Artificial Intelligence to Revolutionize Healthcare: A Comprehensive Analysis," *SciWaveBulletin*, vol. 1, no. 1, pp. 12–19, 2023

N. Firoz, O. G. Beresteneva, A. S. Vladimirovich, **Tahsin, Mohammad Sadman**, and F. Tafannum, "Automated Text-based Depression Detection using Hybrid ConvLSTM and Bi-LSTM Model," in *2023 Third International Conference on Artificial Intelligence and Smart Energy (ICAIS)*, 2023, pp. 734–740, doi: 10.1109/ICAIS56108.2023.10073683

2022

Tahsin, Mohammad Sadman, N. Firoz, S. A. H. Chowdhury, S. Wasei, M. A. Karim, and M. Y. Ara, "Investigation of Credit Card Fraud detection under the lens of Comparative Machine Learning models," in *2022 International Conference on Innovation and Intelligence for Informatics, Computing, and Technologies (3ICT)*, 2022, pp. 25–31, doi: 10.1109/3ICT56508.2022.9990897

M. Islam, **Tahsin, Mohammad Sadman**, F. Alam, S. S. Hossain, A. Deb, and S. Kar, "Understanding Convolutional Neural Network's behavior for Alzheimer's disease on MRI," in *2022 IEEE 13th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)*, 2022, pp. 0179–0185, doi: 10.1109/IEMCON56893.2022.9946529

M. Al Karim, **Tahsin, Mohammad Sadman**, M. U. Ahmed, M. Y. Ara, and S. A. Hasnat Chowdhury, "Evaluating the Performance of ID3 Method to Analyze and Predict Students' Performance in Online Platforms," in *2022 International Conference on Computer Science and Software Engineering (CSASE)*, 2022, pp. 60–64, doi: 10.1109/CSASE51777.2022.9759650

M. U. Ahmed, M. A. Karim, **Tahsin, Mohammad Sadman**, Y. Rahman, and F. Tafannum, "Analyzing Co2 Emission in Developing Countries Using Auto Regression and Auto Regression Walk Forward: A Time Series Approach," in *2022 IEEE Delhi Section Conference (DELCON)*, 2022, pp. 1–5, doi: 10.1109/DELCON54057.2022.9752807

Tahsin, Mohammad Sadman, M. Jobayer, M. B. U. Antor, M. Islam, F. F. Raisa, and M. A. H. Shaikat, "Predictive Analysis & Brief Study of Early-Stage Diabetes Using Multiple Classifier Models," in *2022 IEEE 12th Annual Computing and Communication Workshop and Conference (CCWC)*, 2022, pp. 0203–0207, doi: 10.1109/CCWC54503.2022.9720736

2021

Tahsin, Mohammad Sadman, M. Y. Aziz, T. A. Kabbo, T. Tahsin, N. h. Zumme, and M. I. Hossain, "Data Security Model Using Deep Learning and Edge Computing for Internet of Things (IoT) in Smart City," in *2021 19th OITS International Conference on Information Technology (OCIT)*, 2021, pp. 381–386, doi: 10.1109/OCIT53463.2021.00081

Tahsin, Mohammad Sadman, M. A. Karim, M. U. Ahmed, Y. Rahman, F. Tafannum, and S. Abdullah, "Comparative Analysis of Weather Prediction Using Ensemble Learning Models and Neural Network," in *2021 19th OITS International Conference on Information Technology (OCIT)*, 2021, pp. 325–330, doi: 10.1109/OCIT53463.2021.00071

Professional Experience

Credentialing Analyst | Radiant Data Systems Ltd., Dhaka Mar.–Oct. 2024

- Verified documents and coordinated with U.S. clients to resolve missing information and complete daily operational tasks.

Product Management Intern | REDX, ShopUp, Dhaka Feb.–June 2022

- Contributed to Google Maps and GPS-based fleet work using real-time traffic information for approximately 100 trucks.

Technical Skills

Programming: Python | ML and data tools: TensorFlow, Keras, scikit-learn, NumPy, pandas, Weka | Methods: explainable AI, time-series analysis, data mining, data visualization

Honors and Awards

- Doctoral Fellowship, Texas A&M University-Kingsville, Fall 2026–Present
- Dean's Graduate Merit Scholarship, Texas A&M University-Kingsville
- Research Assistantship Grant, Texas A&M University-Kingsville, Fall 2025
- Vice Chancellor's List, BRAC University, Summer 2020

Professional Service

Peer reviewer: IEEE Access, Heliyon, Biomedical Signal Processing and Control, Intelligence-Based Medicine, Primary Care Diabetes, Systems and Soft Computing, Natural Hazards Research, Franklin Open, Expert Systems with Application.

Technical Program Committee: IEEE ICCE 2026.